

# Microbial detoxification of Pyrethroid insecticide Bifenthrin by selected Fungal strains and Optimizing conditions using Response Surface Methodology

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## Research Article

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# Abstract

Bifenthrin is broad spectrum type I pyrethroid insecticide, used broadly for agricultural and urban purposes, yet very little is reported about its biodegradation. Five fungal strains named *Aspergillus niger* (Asp-N), *Aspergillus flavus* (Asp-F), *Aspergillus fumigatus* (Asp-Fu), *Penicillium chrysogenum* (PC) and *Lasiodiplodia theobromae* (LD) were utilized to degrade bifenthrin for 35 days incubation time. PC is found to be extremely effective in degrading Bifenthrin up to 85%. Furthermore, response surface methodology (RSM) with Box-Behnken design is applied to optimize the degradation conditions with varying pH, temperature (°C) and incubation time (days). The response surface design and analysis of variance depicted the p-value below 0.05 indicating the significance of reaction parameters. The optimal ranges for degrading BF (10 mg/L) by PC were determined to be 30 °C, pH-7 and 24-days incubation period. This fungal strain co-metabolized BF in eutrophic condition within glucose rich medium. The PC degrade BF by hydrolytic cleavage of carboxyl-ester linkage as chromatogram obtained by GCMS with four metabolites. This research provides new insights into the biodegradation of BF by strain *Penicillium chrysogenum* (PC).

## 1. Introduction

Chemically synthesized pyrethroids (SP's) are analogous of Pyrethrins, which are chemical derivative of *Chrysanthemum cinerariaefolium* flowers (Laskowski 2002). SPs are considered as extensively used insecticides throughout the world in the past two decades (Kuivila et al. 2012), especially with the restricted use of organophosphate insecticide in most countries. It is considered as suitable alternative to many extremely toxic chemicals and its usage as well as production is thought to be increase in coming few years. Presently, it is the leading insecticide (Weston et al. 2009) contributive to > 25% of the global insecticide retails sales to consumers (Zhang et al. 2010). Pyrethroids contain cyclic carboxylic acids with aromatic alcohols linkage through an ester bond.

Due to its easy availability, pyrethroids are extensively used throughout the main domains of household and commercial purposes. Pyrethroids are reported to be present in all the urban dwells and sediments in addition to presence of residues in aquatic animals (Riaz et al. 2018; Ariyani et al. 2020; Holmes et al. 2008; Weston et al. 2011). Its excessive use raises a prominent public concern related to its potential toxicity in human population due to presence of its residues in food products (Prodhan et al. 2021). Plentiful studies have shown that SPs may cause neurotoxicity leading to nerve damage (Amin et al. 2021) reproductive system damage (Wang et al. 2020) and genotoxicity with hepatorenal damage (Mohafresh et al. 2021) in various nontarget organisms. In view of these findings, it is critically important to develop cost effective strategies and solutions to remove SPs from agricultural products and environmental surroundings.

One of the most extensively used and popular SPs is Bifenthrin[2-methylbiphenyl-3-ylmethyl-(Z) -(1RS)-3-(2-chloro-3,3,3-trifluoroprop-1-enyl)-2,2-dimethylcyclopropane carboxylate] which is considered competent against broad spectrum pests in crops as well as resident termites (Wang et al. 2009; VanDusen et al.

2016). BF half-lives ranges from 14-days minimum to maximum 1 year in general however it can be different in variable moisture, soil type, temperature and other environmental conditions (Mohapatra and Ahuja 2009; Mukherjee et al. 2020; Ramasubramanian and Paramasivam 2021). World Health Organization has categorized BF as class II toxic moderately hazardous chemical (WHO 2020). BF used in different territories may release to environment either by surface runoff, volatilization or leaching. It undergoes various environmental phenomena relating to its degradation such as photodegradation, hydrolytic decomposition and soil microbial decontamination (Gajendiran & Abraham 2018).

Microbial consortium is considered safe and effective biodegradation for SPs (Bhatt et al. 2019a). Many studies are present in which bacterial genera from *Pseudomonas*, *Achromobacter*, *Clostridium*, *Acidomonas*, *Streptomyces*, *Bacillus*, *Serratia* and genus *Lysinibacillus* strains have been studied for SPs detoxification (Cyon and Piotrowska-Seget 2016; Zhao et al. 2019; Birololi et al. 2019; Hu et al. 2019). Fungus is also proven to be effective against SPs insecticides. Many fungal strains have found to be more degradative than bacteria (Bhatt 2019b; Gangola et al. 2019). Pyrethroid biodeterioration is reported by only a few fungal strains like *Hansenula anomala* (Chen et al. 2012a), *Aspergillus terricola*, *Phaenerochaete chrysosporium* and *Cladosporium* sp. (Liu et al. 2021; Saikia and Gopal 2004; Deng et al. 2015; Chen et al. 2011). Different metabolic pathways are predicted for SPs biodegradation by fungi based on result obtained from various characterization techniques. Some intermediate metabolites are common among any SPs such as 3-phenoxybenzoic acid is common in Fenpropathrin and BF for microbial degradation. However, very little information is available in literature about the BF degrading agents.

In the recent study, five different fungal strains PC, Asp-F, Asp-Fu, LD and Asp-N were grown and used to degrade BF. The key objective of this research is to optimize the degradation parameters by response surface methodology (RSM), predict its degradative route and investigate the potential of different fungal strains to remediate the BF-contaminated environment. Gas chromatography-mass spectrometry (GCMS) and Uv-visible spectrophotometer were used as analytical techniques to analyze the degradation products in varying reaction conditions. Finally, data obtained in the study might be able to predict the varying capability of different fungal strains in BF bioremediation.

## 2. Materials And Methods

### 2.1. Chemicals and Media

Analytical standard pyrethroid insecticide BF ( $C_{23}H_{22}ClF_3O_2$ ) (with a purity of 99.9%) was purchased from Dr. Ehrenstorfer company, Germany while Dichloromethane (DCM;  $CH_2Cl_2$ ) and anhyd. Sodium Sulphate ( $Na_2SO_4$ ) were obtained from RCI Lab-scan, Thailand southeast country. Distilled water and ethanol were used to prepare all solutions and apparatus sterilization. All the other chemicals used in the experiment were obtained from Merck, Germany.

Potato dextrose agar (PDA) as fungal culturing medium and potato dextrose broth (PDB) were used for BF degradative fungal cultivation and growth. After various trial-based experiments, 7.2 was set as final optimum growth pH. Both growth media were autoclaved separately, at 121 °C for 30 min.

## **2.2. Fungal strains inoculum preparation**

Fungal inoculum of five strains Asp-F, LD, PC, Asp-N, and Asp-Fu were kindly provided by laboratory of Mycotoxicology, Environmental Sciences department, Fatima Jinnah Women University, Pakistan. Each fungal strain was cultured and spread in autoclaved-sterilized petri-dishes having a rich potato dextrose agar (PDA) medium. It was wrapped with parafilm masking tape and kept in incubator at 29.5°C. Each step of experimentation was conducted in a laminar flow cabinet to guarantee ample sterilization. After growth of a week, a sterilized surgical blade was used to relocate/transfer the fungal mycelia into an Erlenmeyer flask having 1/3rd of distilled water. Transfer was also performed in laminar flow cabinet. Complete blending and mixing are done by employing each flask on vortex (Ahmad 2020).

## **2.3. Bifenthrin biodegradation in relation to pH and temperature variation by response surface methodology (RSM)**

To assess the degradation capacity of selected fungal strains in Bifenthrin transformation, biodegradation assays were arranged. The biodegradation experiment was achieved on BF aimed at bioremediation by fungal cultures in PDA and duplicates for each sample were prepared also. All the prepared flasks were placed in incubator at  $29.5\text{ }^{\circ}\text{C} \pm 2$  and extracted every first day of week. All the degradative assemblage is performed in duplicates and prepared in Erlenmeyer flask. In further experiment, the strain which resulted in maximum removal efficiency of BF, were subjected to RSM for optimizing the response of degradation by various alterations in experimental conditions (Bhatt et al.2020b). To assess the effect of temperature, experimental setup was placed at 20, 25, and 30 °C. The pH was adjusted at 5, 7 and 9 to assess the degradation potential of selected strain. For all experimental conditions, control setup (uninoculated flask) was also organized without any fungal strain.

## **2.4. Analysis of Bifenthrin degrading metabolites by UV-Vis and GC-MS**

Samples were extracted after every eight days to assess both quantitatively and qualitatively by UV-Vis spectrophotometer and GC-MS. The extraction was obtained by extracting funnel after adding 5 + 5 mL dichloromethane as extracting solvent. The solution was vigorously shaken for few minutes and allowed to stand still. To remove moisture from the extracted layer, sodium sulphate anhydrous was added and set aside (Zheng et al. 2016). The extractant was heated at hot plate to vaporize till 2 mL which was stored at temperature of -4°C in refrigerator until further investigation by GC-MS.

## **3. Kinetic Investigation**

The BF biodegradation parameters were analyzed according to first-order reaction kinetics (Bhatt et al. 2020a; Pankaj et al. 2016; Chen et al. 2013; Cycon et al. 2014).

$$C_t = C_0e^{-kt}$$

1  
.....

Whereas, C<sub>0</sub> is the initial concentration of BF, C<sub>t</sub> is the BF concentration at 8, 16-, 24-, 32-, and 40-days extraction; k is rate constant, and t depicted the incubation time (days).

BF Half-life was enumerated as follows;

$$T1/2 = ln\frac{2}{k}$$

2  
.....

Degradation percentage is calculated by applying the following equation;

Percentage of degradation = (A<sub>0</sub> - A<sub>e</sub> / A<sub>0</sub>) ..... (3)

### 3.1. Experimental design and statistical investigation

Five types of fungal strains were subjected to degrade BF for four weeks. Maximum degradation percentage were deciding parameter for the selection of fungal strain subjected to RSM. Experimental setup was designed as per rules of RSM. Box Behnken design (BBD) was adopted to analyze the effects of variable conditions on degradation efficiency (Hamad 2020). BBD of three levels (-1, 0, 1) for experimental parameters were applied for experiment, including pH, time (days) and temperature (°C). These applied parameters are coded as A = pH, B = Time and C = Temperature as shown in Table 2. Applied parameters were analyzed rendering to following Eq. (4):

$$Y_i = (X_i - X_o) l\Delta X_i$$

4  
.....

Whereas, Y<sub>i</sub> is dimensionless-independent variable value, X<sub>i</sub> is the real-independent variable value, X<sub>o</sub> is value for independent variable value at midpoint and ΔX<sub>i</sub> is change in variable value.

Response surface plots, contour graphs and ANOVA were enumerated by using Minitab-18 for explaining degradation efficiency of selected strain.

## 4. Results And Discussion

### 4.1. Rate of BF degradation by fungal strains

In the present study, five unique fungal strains were used to degrade BF in liquid medium. The overall pH of all sample solutions was kept neutral to avoid any degradation due to change in pH. The maximum degradation of 85.4% was observed by *Penicillium Chrysogenum* with half-life of 9.7 days as shown in Table 1. Among three fungal strains of genus *Aspergillus*, Asp. F depicted the maximum degradation (*i.e.*, 55%) with half-life of 25.9 days. While lowest degradation of 28% was observed in Asp. Fu with half-life of 58.2 days. Control sample (devoid of any fungal strain) depicted 5% degradation only, which might be attributed as BF hydrolysis in the presence of extra carbon source (PDB medium) as shown in Table 1. Log plots were designed after applying the first order reaction kinetics to the degradation percentage (Fig. 1). Half-life and degradation constant were also intended by log plots. Descending order of half-lives of used strains is as follows;

Asp.Fu > Asp. N > LD > Asp. F > PC

The Ap. N and Asp-Fu depicted the 35% and 28% degradation respectively, with neutral pH in four weeks while Saikia and Gopal conducted a similar pyrethroid degradation study, where Asp. N depicted only 10% degradation in acidic pH, above 20 days incubation (Saikia and Gopal 2004). Fareed et al. (2017) proposed that Asp-N fungal strain can remove diazinon to 46% within 9 days of incubation at 30°C. Many researchers explained that the fungal strains related to genus *Aspergillus* order Eurotiales, family Trichocomaceae depicts high degradation capacity for the organophosphate-insecticides (having inorganic phosphorus) and other pesticides (Oliveira et al. 2015). Asp. N have myo-inositol-hexa-kis-phosphate phosphor-hydrolase enzyme which can remove inorganic phosphorus (Shah et al. 2017). BF ( $C_{23}H_{22}ClF_3O_2$ ) doesn't have any phosphorous so it might be the reason of its comparatively less degradation by three strains of *Aspergillus*.

The sample with PC depicted the lowest half-life (9 day) with rate of highest degradation (85.4%). Shan and co-workers depicted the efficient reduction in BF residue in tea by inoculation with PC. BF is almost 90% removed by one of the strains of *C. pelliculosa* within five days incubation (Chen et al. 2012b).

Uninoculated sample (control) displayed the highest half-life (154 days) with minimal degradation (5%) that can only be attributed as hydrolysis by liquid medium.

Table 1

Fungal degradation percentage along half-life of BF attained for different fungal strains by UV–Visible spectrophotometer

| Sr. No | Fungal strains              | Rate constant ( <i>k</i> ) | BF half-life (in Days) ( <i>t</i> <sub>1/2</sub> ) | Percentage degradation (%) |
|--------|-----------------------------|----------------------------|----------------------------------------------------|----------------------------|
| 1      | Asp. N                      | 0.0131                     | 53                                                 | 35                         |
| 2      | Asp. F                      | 0.0267                     | 25.9                                               | 55                         |
| 3      | Asp. Fu                     | 0.0119                     | 58.2                                               | 28                         |
| 4      | PC                          | 0.0709                     | 9.77                                               | 85.4                       |
| 5      | LD                          | 0.0213                     | 32                                                 | 49                         |
| 6      | Control (inoculated) sample | 0.0045                     | 154                                                | 5                          |

## 4.2. Optimization of Bifenthrin degradation conditions in PC by RSM

In further experiment, PC was employed to optimize the response of degradation by varying pH (A), time (B) and temperature (C) (Table 2). Strain PC effectively degraded BF at different temperatures and pH conditions. RSM with Box-Behnken design was employed to optimize the different factors in PC inoculated sample setup (Table 3) (Fig. 3). The experiment was designed according to the given matrix in table no. 2. The data obtained from 15 experiments were employed to analyze with Minitab-17. The deduced outcomes of fitted equation and analysis of variance for response surface design (ANOVA) of the running experiment is given in Table 4. Experimental outcomes of BF were fitted in the given quadratic polynomial equation;

### Regression Equation in Uncoded Units

$$\begin{aligned} \text{Degradation Efficiency} &= 76.00 - 0.625 A + 2.938 B + 3.563 C - 21.94 A^*A - \\ &0.81 B^*B + 0.94 C^*C - 0.75 A^*B + 0.00 A^*C \\ &+ 1.13 B^*C \end{aligned}$$

**Table 2** Experimental factors and ranges of independent factors according to design of RSM

| Experimental factors/parameters | Symbols | Levels |    |    |
|---------------------------------|---------|--------|----|----|
|                                 |         | -1     | 0  | 1  |
| pH                              | A       | 5      | 7  | 9  |
| Time (Days)                     | B       | 8      | 16 | 24 |
| Temperature (°C)                | C       | 20     | 25 | 30 |

**Table 3** Box-Behnken response surface experimental design and values/response of dependent variable for BF degradation by PC

| Run | A  | B  | C  | X    |
|-----|----|----|----|------|
| 1   | 1  | 0  | 1  | 57.0 |
| 2   | 1  | -1 | 0  | 52.0 |
| 3   | -1 | 0  | 1  | 58.0 |
| 4   | 0  | -1 | 1  | 75.0 |
| 5   | 0  | -1 | -1 | 68.0 |
| 6   | 0  | 1  | -1 | 75.0 |
| 7   | 1  | 0  | -1 | 52.0 |
| 8   | -1 | 0  | -1 | 53.0 |
| 9   | 0  | 1  | 1  | 86.5 |
| 10  | 0  | 0  | 0  | 76.0 |
| 11  | -1 | 1  | 0  | 56.0 |
| 12  | 0  | 0  | 0  | 75.0 |
| 13  | 1  | 1  | 0  | 53.0 |
| 14  | 0  | 0  | 0  | 77.0 |
| 15  | -1 | -1 | 0  | 52.0 |



Table 4 depicted the analysis of variance (ANOVA) to fit in the above quadratic-polynomial equation. Significant p-values (0.001) of model suggested it as a best fit to explain BF degradation by PC. The linear p-values for time and temperature are less than 0.05 ( $p < 0.05$ ) indicating their efficient effect on BF degradation while linear and square value of pH is insignificant ( $p > 0.05$ ). ANOVA revealed that no significant two-way interaction was present between effects of three parameters. The p values for independent variables A, B and C were obtained as 0.5, 0.02 and 0.01 respectively. The interaction of pH, time and temperature is observed to be insignificant as their p-value is greater than the significance level. The  $R^2$  value of results showed that model depicted 98.32% variance in degradation efficiency, proved the model best fitted for the given data. The normal probability graph of BF (as shown in figure 2) followed a straight line and showed the normal distribution of residuals. Residual verses fit plot showed the random distribution of points across zero line hence supported constant variance of residuals. In residual verses order plot, points fall randomly across the line which proved used variables as independent. Surface plots and contour plots are designed to analyze the effect of two parameters while considering the third variable as constant. Surface plots (figure 3) depicted the 3D graph for a functional relationship between degradation efficiency and factors A, B C. The peaks and valleys showed the maximum and minimum response values in surface plots respectively. Contour plots are companion plots to surface plots as shown in figure 4. A two-dimensional view is provided by contour plot in which all values of variables having same degradation efficiency are connected by contour lines. Darker areas showed maximum response. These higher efficiency values are seemed to form a central ridge which is located in middle of graph while stripes in extreme right and left of graph represent pH-time-temperature combinations that results in low degradation efficiency (>55% to 65%).

Box-cox plot for the degradation efficiency of PC in various runs is obtained by Minitab as shown in Fig. 5. 95% confidence level for  $\lambda$  is obtained with 2.15 upper CL (*i.e.*, upper confidence limit). The estimated  $\lambda$  value came out to be 1 which corresponds to no transformation of original data. Bubble plots represents the degradation efficiency of BF by size of bubbles as showed in Fig. 5. Degradation efficiency is showed by size of each bubble.

Response optimization plot is shown in Fig. 6. Maximum current degradation is obtained when A is equal to 0 (pH = 7) while B and C is equal to 1 (*i.e.*, 1 is coded for temperature of 30 °C and 24-days incubation period).

**Table No 4**

Analysis of Variance ANOVA for response surface design

|                   | DF | Adj SS  | Adj MS  | F-Value | P-Value |
|-------------------|----|---------|---------|---------|---------|
| Model             | 9  | 1985.30 | 220.59  | 32.50   | 0.001   |
| Linear            | 3  | 173.69  | 57.90   | 8.53    | 0.021   |
| A                 | 1  | 3.12    | 3.12    | 0.46    | 0.528   |
| B                 | 1  | 69.03   | 69.03   | 10.17   | 0.024   |
| C                 | 1  | 101.53  | 101.53  | 14.96   | 0.012   |
| Square            | 3  | 1804.30 | 601.43  | 88.61   | 0.000   |
| A*A               | 1  | 1776.94 | 1776.94 | 261.80  | 0.000   |
| B*B               | 1  | 2.44    | 2.44    | 0.36    | 0.575   |
| C*C               | 1  | 3.25    | 3.25    | 0.48    | 0.520   |
| 2-Way Interaction | 3  | 7.31    | 2.44    | 0.36    | 0.786   |
| A*B               | 1  | 2.25    | 2.25    | 0.33    | 0.590   |
| A*C               | 1  | 0.00    | 0.00    | 0.00    | 1.000   |
| B*C               | 1  | 5.06    | 5.06    | 0.75    | 0.427   |
| Error             | 5  | 33.94   | 6.79    |         |         |
| Lack-of-Fit       | 3  | 31.94   | 10.65   | 10.65   | 0.087   |
| Pure Error        | 2  | 2.00    | 1.00    |         |         |
| Total             | 14 | 2019.23 |         |         |         |

### 4.3. Biodegradation pattern and kinetics of BF by *P. chrysogenum*

To identify the degradation products of BF by PC, fungal cell free-filtrates were extracted and analyzed. The biodegradation metabolic products of BF at 10mg/L concentration were scrutinized every 8th day by GC-MS for 24 days duration. The degradation products of BF identified by MS is matched by authentic standards of NIST library database.

The GC-MS spectrum of first extraction depicted two major peaks (Fig. 8). Peak at 22.90 min RT depicted the presence of BF. The intensity of this peak decreases over time and became almost invisible at the end of fourth extraction. BF degradation is observed concomitantly with formation of five other metabolites. The BF metabolites showed in table 5, corresponded to (A) 2, methyl-3-biphenyl methanol (RT = 8.9 /min),

(B) 2, chloro-6-Fluorobenzyl alcohol (RT = 11.1 /min), (C) 3, Phenoxy benzene methanol (RT = 18.2 /min), (D) 1,2 benzene dicarboxylic acid (RT = 20.5 /min).

On the basis of the above stated metabolites, degradation pathway for BF was predicted. The degrading parameters of time, pH and temperature are the three influencing factors which strongly effect the degradation of BF by various micro-organisms. Our study depicted the efficient degradation of BF at a wide range of temperature (20, 25, and 30 °C) and time (8,16 and 24 days). These factors are considered very important for microorganisms to be employed for bioremediation of xenobiotics (Singh et al. 2006). Moreover, the optimal conditions for BF degradation are 30<sup>0</sup>C, pH 7 and 24-days incubation time *via* RSM. Under these optimal conditions, fungal strain *P. chrysogenum* degraded BF (10mgL<sup>-1</sup>) upto 86.5%. The contour plot and box-cox plots confirmed the effective degradation of BF at optimized conditions. The resembling results to this study were previously reported on Allethrin degradation by *Fusarium proliferatum* (Bhatt et al. 2020b). RSM is considered as an efficient statistical technique, able to optimize the degradation conditions of various microorganisms. It is an inexpensive, faster and efficient method for optimizing factors and gathering research results in an effective manner (Zhang et al. 2010; Chen et al. 2011; Chen et al. 2012c; Drista 2009).

BF is co-metabolized in the presence of extra carbon source (PDB medium) by *P. chrysogenum* in the given study. It depicted the adaptation of BF in only eutrophic conditions. However, this research did not quite agree with Chen et al (Chen et al. 2012b) observations, who reported BF degradation by *C. pelliculosa* strain ZS-02 in both oligotrophic and eutrophic conditions. Our study showed efficient degradation of BF in the presence of glucose which might be attributed as the extra carbon source for co-metabolism of BF enhanced degradation. Similarly, efficient and enhanced degradation of organophosphate pesticides by different microorganisms in eutrophic conditions was reported by Cycon et al.2009 and Anwar et al. 2009.

Temperature variation is an important factor in biodegradation of pesticide. Extremely low and high temperatures can destroy the microbial communities which in turn decrease the BF degradation. In our study, gradual increase in temperature enhances the degradation and maximum degradation is observed at 30°C. Zhang and co-workers depicted the maximum BF and chlorpyrifos degradation at 30°C and pH 7. Chen et al. reported the optimum BF degradation by *C. Pelliculosa* strain at 32.3°C.

The pH of culturing medium along incubation time for fungal strain growth strongly influences the PC degradation efficiency. Slightly low BF degradation was achieved at pH 5 owing to the acidic activity of medium which supports higher persistent. However, as inoculation/incubation time increases, the difference in BF degradation at varying pH level diminishes. Wang et al. also reported the highest degradation of chlorpyrifos at pH 7. It showed that the both acidic and basic pH is highly detrimental to existence of microbial community leading to decrease in recalcitrant compound degradation (Song et al. 2014). The degradation of BF to such higher extent is not only a consequence of fungal incorporation but also a result of chemical hydrolysis. The hydrolytic potential of water increases to 10-fold for every point of pH change (Deer and Beard 2001). Meena and coworkers reported the increased dissipation of BF at

pH 7 as compared to pH 4. However further increase in pH with incubation time diminishes the sole effect of pH in BF degradation as demonstrated by statistical analysis.

The BF metabolize by hydrolytic cleavage of carboxyl-ester linkage to produce 2-methyl-3-biphenyl-methanol. Moreover, 2-methyl-3-biphenyl-methanol was subsequently cleaved at biphenyl to form 2, chloro-6-Flouro benzyl alcohol and 3, Phenoxy benzene methanol resulting in its metabolite formation (Fig. 7). Previous studies have also confirmed that the hydrolytic cleavage is the major step during pyrethroid degradation (Zhan et al. 2020). Moreover, degradation rate is highly dependent on molecular structure of pyrethroid as explained by Chen et al. Various pyrethroids including cyfluthrin, fenvalerate, cypermethrin, bifenthrin and deltamethrin were dissipated at much faster rate than fenpropathrin proved that with the presence of any halide moiety (i.e., F, Br, Cl group) at chrysanthemic acid, degradation efficiency increases. It might be because of the increased hydrolase activity in presence of these groups.

**Table No. 5:BF metabolites exhibited by GCMS obtained from PC degradation**

| Codes | RT (Retention time/minutes) | Compounds                         |
|-------|-----------------------------|-----------------------------------|
| A     | 22.9                        | Bifenthrin                        |
| B     | 8.9                         | 2, methyl-3-biphenyl methanol     |
| C     | 11.1                        | 2, chloro-6-Flouro benzyl alcohol |
| D     | 18.2                        | 3, Phenoxy benzene methanol       |
| E     | 20.5                        | 1,2 benzene dicarboxylic acid     |

## 5. Conclusion

BF microbial degradation by five different fungal strains were studied to enumerate their potential for pesticide degradation. PC found to be very effective for BF degradation to an extent of 85.4%. Furthermore, response surface methodology was used to optimize the degradation response of BF by PC. Maximum degradation (*i.e.*, 89.5%) is observed at 30 °C, pH-8 and 24 days incubation time. Fungal degradation and chemical hydrolytic cleavage of the ester bond is responsible for the metabolites production of BF. BF degraded to produce 2-methyl-3-biphenylmethanol and 1,2-Benzenedicarboxylic acid. Furthermore, 2-methyl-3-biphenylmethanol was subsequently cleaved at biphenyl to form 2, chloro-6-Flouro benzyl alcohol. We can conclude that PC has able to deteriorate BF with lowest cost and greater efficiency. However, enzymatic assay is required to further understand the mode of deterioration of BF by PC.

## Declarations

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## Data Statement

Data will be available on demand.

## Ethical approval

The current work does not involve experimentation on any human or animals.

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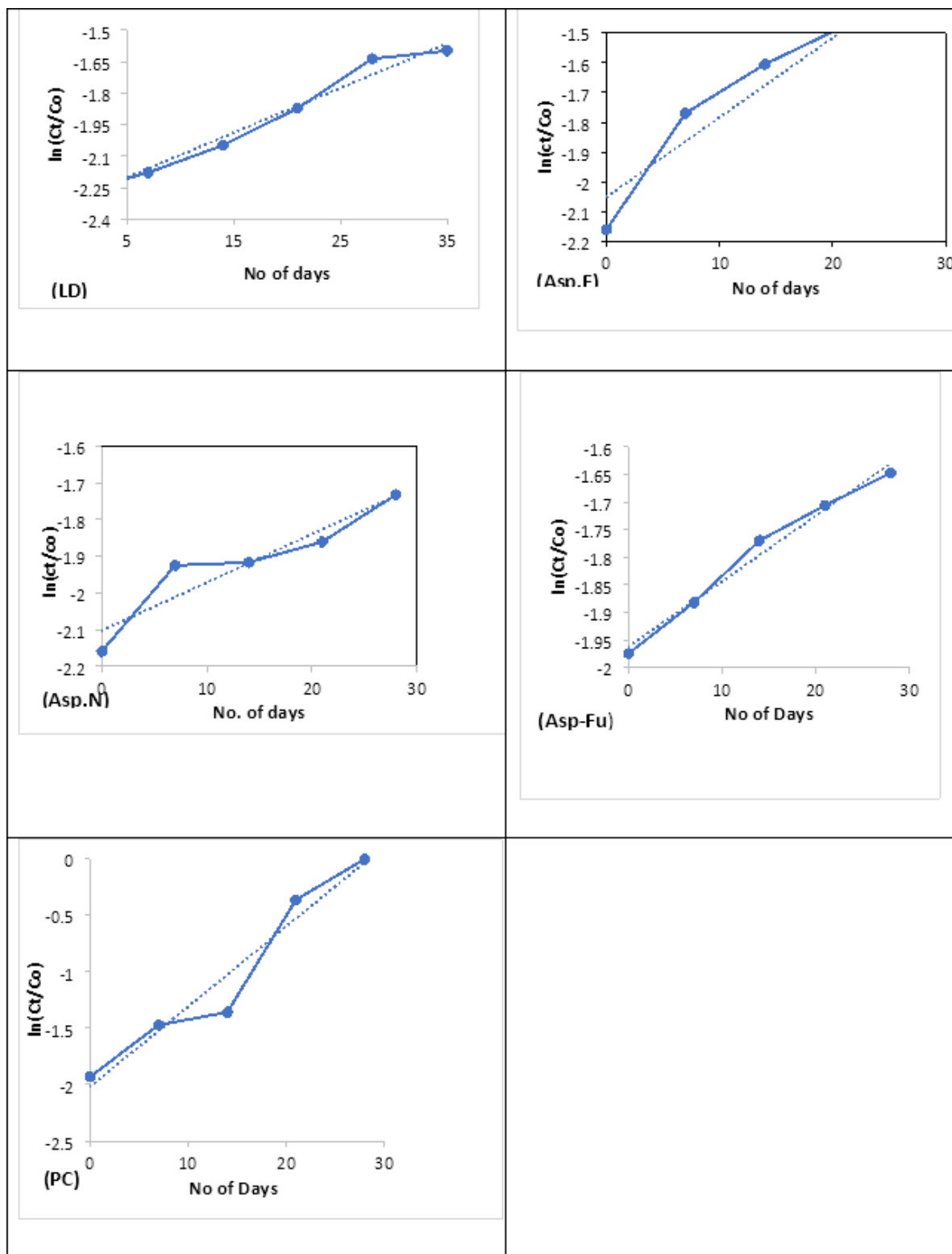
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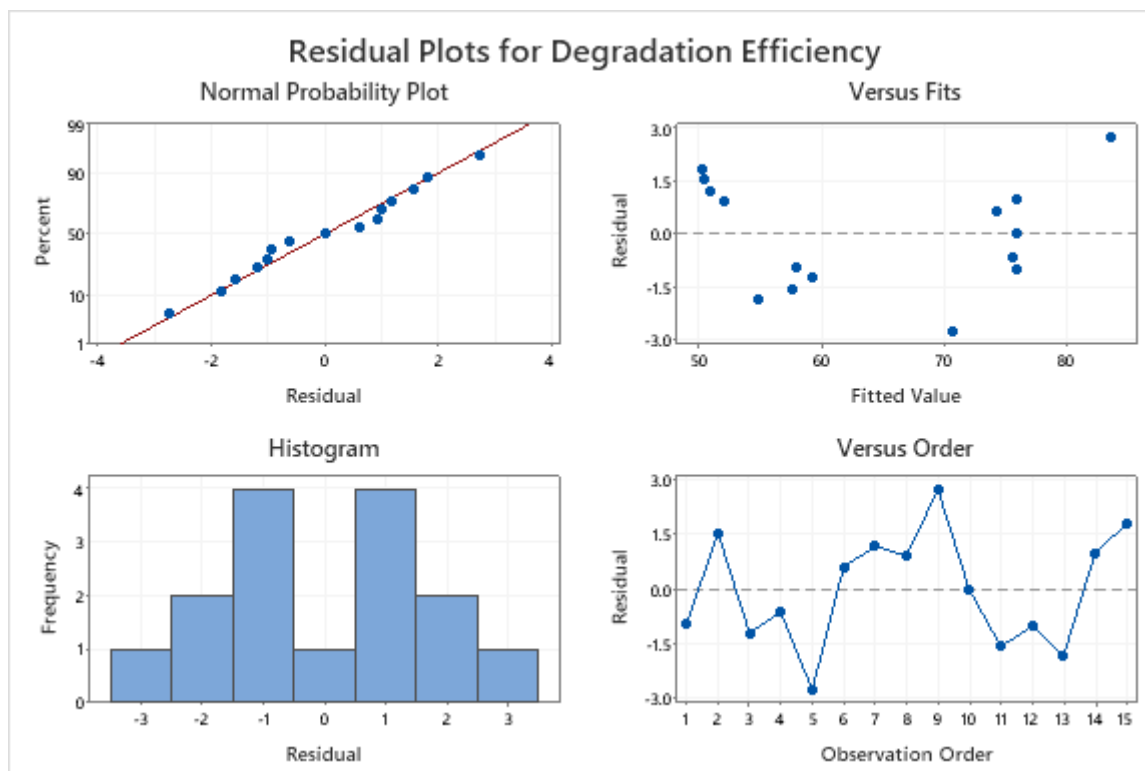
## Figures





**Figure 1**

Log plots and slope findings of five fungal strains utilized for BF degradation in one month incubation time



**Figure 2**

Two-way analysis of variance (ANOVA) plots for degradation efficiency of PC

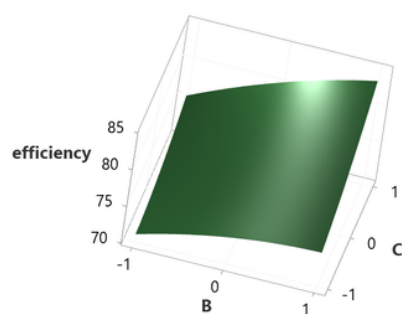
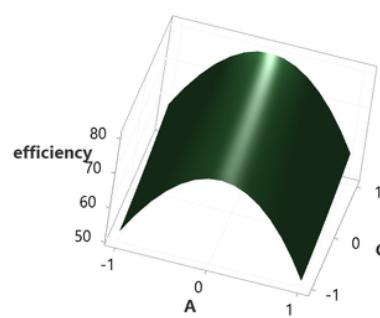
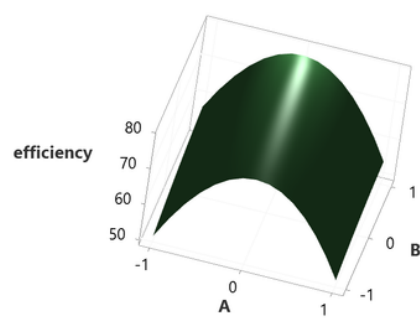
## Surface Plots of Degradation Efficiency

Hold Values

A 0

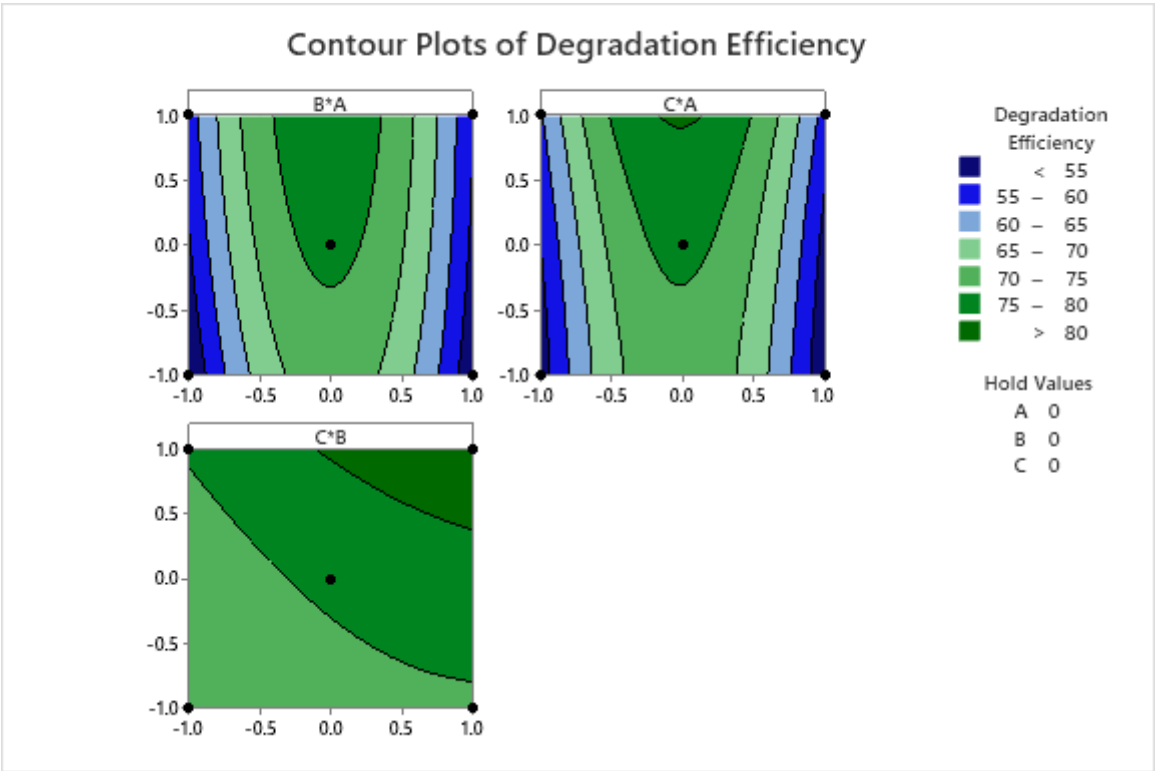
B 0

C 0



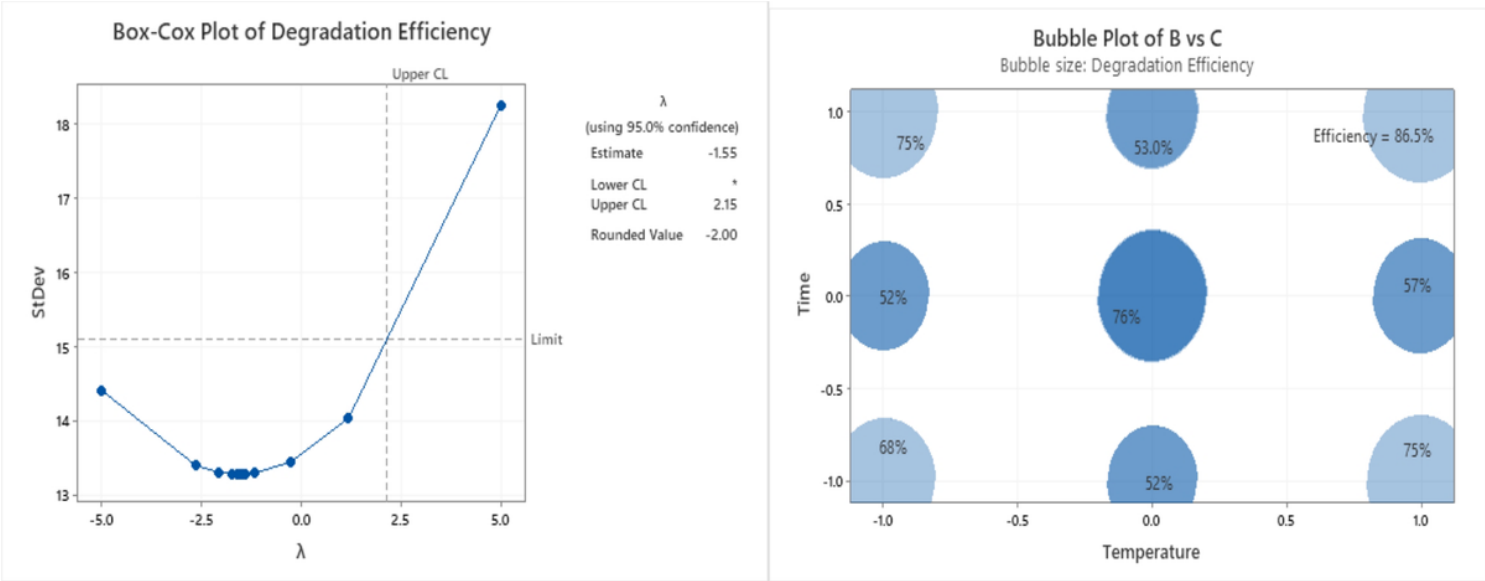
**Figure 3**

Surface plots for degradation efficiency of PC(a) Time\*pH (b) Temperature\*pH (c) Temperature\*Time



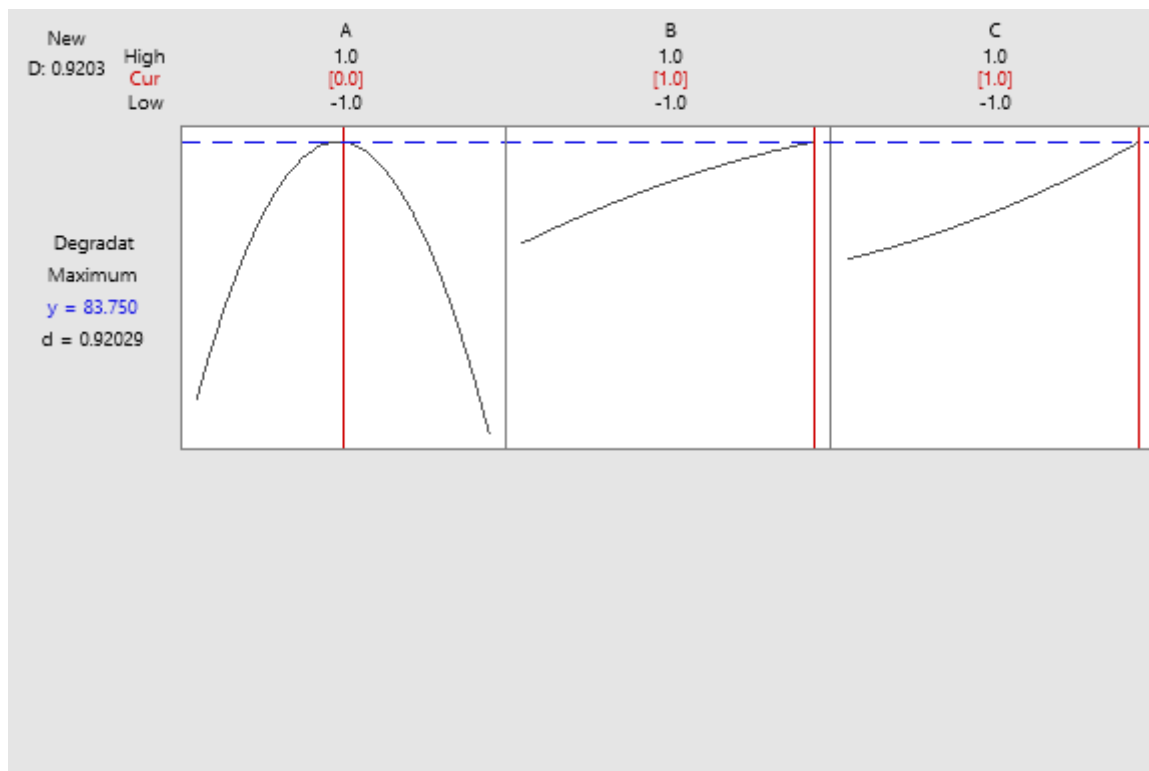
**Figure 4**

Contour plots for degradation efficiency of PC between factors pH, time and temperature Experimental plan and response values for various runs of BF degradation efficiency are present in table 3.



**Figure 5**

A Box–Cox plot and bubble plot (showing degradation efficiency inside bubble) for the correlation-coefficients of several optimized conditions



**Figure 6**

Response optimization of BF degradation efficiency by *P. Chrysogenum* under varying conditions of pH(A), Time(B) and Temperature(C)

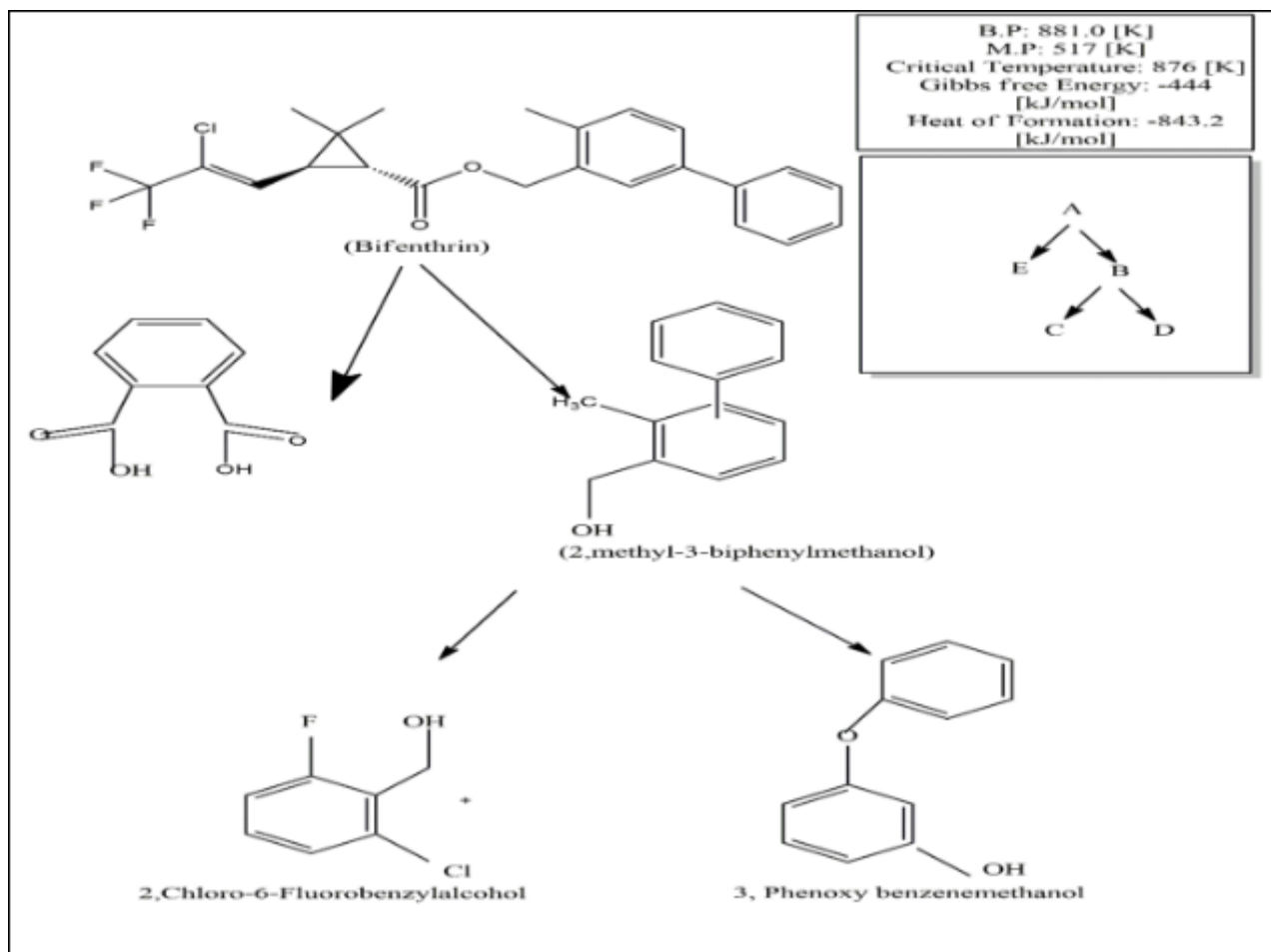
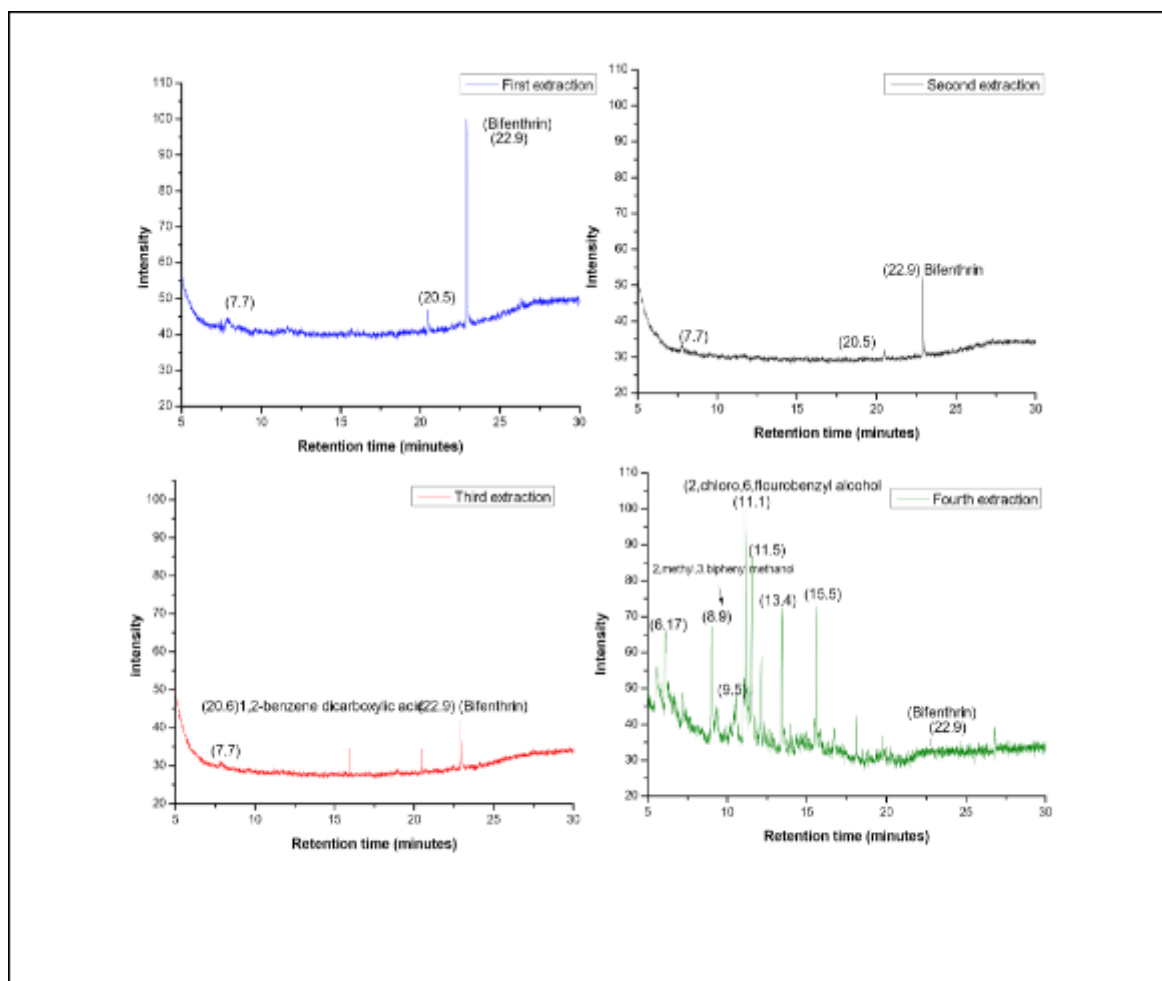


Figure 7

Flow sheet diagram of BF degradation by PC



**Figure 8**

GC-MS chromatograms of BF degradation by PC fungal strain at optimized conditions of 30 °C and pH 8

## Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [GraphicalAbstract.png](#)